

ELI MK-XI — Full Project Audit (Engineering Reference)

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Purpose: A durable, accurate map of the codebase to be used when **refactoring, upgrading, and fixing**. It records what each subsystem is, where it lives, its entry points, its current status, and every known issue — so future work starts from facts, not guesses.

Snapshot: commit a641471, 2026-06-17. Counts marked (*runtime snapshot*) come from the live EXPLAIN_MEMORY_RUNTIME dump in the 2026-06-17 session and drift over time.

Companion docs: blueprints/adaptive_inference_governor.md (deep-dive on the think/budget/throughput policy), plus the existing blueprints/*.md set (indexed in §13).

Accuracy policy (how to trust this document)

Every statement is one of: - **[V]** verified this session by reading the code / running it (file:line given); - **[L]** evidenced by the 2026-06-17 session log (quoted/derived); - **[B]** sourced from an existing blueprint or the maintainer memory index, **not** re-verified line-by-line this session (flagged so it can be re-checked before relying on it).

If a statement has no tag it is a structural fact from the file tree / wc / grep run this session **[V]**.

1. At a glance (verified this session)

Metric	Value
Python files under eli/	370 (+ <i>api/server.py, the FastAPI web app, ~3,884 lines</i>)
Python LOC under eli/	143,432 (<i>measured 2026-07-01</i>)

Metric	Value
Largest modules	executor_enhanced 14,294 · engine 13,326 · GUI eli_pro_audio_gui_MKI 10,985 · router 7,076 · labs_tab 5,562 · memory 4,415 · deterministic_grounding_gate 4,292 · agent_bus 3,073 · gguf_inference 2,538
Capabilities (capability_manifest.json)	208
SUPPORTED_ACTIONS (executor_enhanced.py)	174
Specialist bus agents	14 (named in §4.4)
Test files / suite	189 files; full run 7,347 passed / 5 failed / 45 skipped / 2 xfailed (2026-07-01)
Test groups	tests/claims (11), tests/regression (6), tests/generated (3)
SQLite stores	4 (user, agent, system_index, coding_memory)
Vector index	FAISS, 1,228 vectors, nomic-embed-text-v1.5.Q4_K_M (<i>runtime snapshot</i>)
Knowledge graph	50 entities / 49 relations (<i>runtime snapshot</i>)
Entry point	python -m eli → eli/__main__.py:main(); engine CognitiveEngine.process at engine.py:8790

Subpackage LOC map (eli/)

Package	files	LOC	Role
execution	14	23,082	router + executor (the action surface)
runtime	79	22,225	grounding gate, guards, con- tracts, remedia- tion — the
gui	6	17,618	catch-all PyQt app + tabs
kernel	8	14,406	engine.py cogni- tion con- troller + state
cognition	27	13,739	agents, infer- ence, persona, reason- ing modes

Package	files	LOC	Role
memory	13	6,936	SQLite + FAISS + KG + recall vision, STT, TTS, screen, gaze hardware, settings, paths, model tier, dag, netguard proactive daemon, jobqueue, sched- uler, world- room self- model, habits, sum- maries frontier coding agent (plan→solve→verify) news fetcher, image engine (im- age_engine nested) logging, con- tracts, plugin host, world avatar empty/dead (see §10)
perception	20	6,821	
core	23	6,073	
planning	24	3,795	
learning	12	3,315	
coding	12	2,031	
tools	4*	1,903	
utils/contracts/plugins/system/world/cli		<1k each	
guards/integrations	1	0	

2. Repository map (top level)

- eli/ — the application (above).
- blueprints/ — design docs (index §13). This audit + the governor plan live here.
- artifacts/ — runtime state: db/ (the 4 SQLite stores), vectors/ (FAISS index+meta), runtime_snapshot.json, runtime_hardware_profile.json, analyze_image_*, eval outputs, world snapshots.

- models/ — GGUF weights (text + embeddings + vision + LoRA); no model is baked into code.
 - tests/ — pytest suite (§11).
 - tools/, scripts/, ops/, packaging/, training/ — eval/build/train/ops.
 - config/ — trusted agents, settings.
 - Install: install.sh / install.bat / install.ps1, requirements*.txt, requirements.lock.txt, pyproject.toml.
-

3. Control flow (the cognition pipeline)

Single entry: `CognitiveEngine.process()` (engine.py:8790). 12 logical stages **[L/V]** (stage map verified live via `EXPLAIN_COGNITION_RUNTIME`):

1. Perceive + ingest — `gui/eli_pro_audio_gui_MKI.py` (text) / `perception/audio_stt.py` (voice).
2. Input normalization + guards — `engine.py` input guards (long-question, command, wake-word).
3. Router + task decomposer — `execution/router_enhanced.py:route()` → {action, args, confidence, meta}.
4. Truth / grounding gate — forbids a direct LLM answer when grounding is required.
5. Executive controller / planner — picks agent profile + order.
6. Agent bus — up to 14 specialist agents on a `ThreadPoolExecutor`, ordered by a dependency DAG.
7. Working-memory / context assembler.
8. **Single inference broker** — `cognition/inference_broker.py`; all model calls serialize on `_LLM_CALL_LOCK` (gguf_inference.py:515). llama.cpp is not concurrency-safe; vision hot-swap and the ambient daemon coordinate on this lock too. **[V]**
9. Reasoning / synthesis — pure chat / grounded / hybrid; reasoning-mode strategies in §7.
10. Output governor — `cognition/output_governor.py` (1,200 LOC) shape/evidence/confidence.
11. Delivery — GUI / TTS (`perception/tts_router.py`, Piper).
12. Learning + state update — `memory/memory.py` stores turns/memories/habits/failures.

Fast path: `_PHASE45_DIRECT_FAST_ACTIONS` (engine.py:78) — deterministic actions whose executor result is authoritative are executed and returned verbatim, bypassing synthesis (e.g. `SCREENSHOT`, `ANALYZE_IMAGE`, `CHECK_JOB`, and now `CREATE_FILE` after a641471). **[V]**

4. Subsystem audit

4.1 Router — `execution/router_enhanced.py` (7076) Deterministic, priority-ordered intent parser: explicit regex routes → `multi_commandprepass` (`runtime/command_splitter.py`) → LLM intent-resolver fallback (`cognition/llm_intent.py`). Public `entryroute(text)`. ****Status:**** the highest-churn correctness surface; regex over/under-capture is the dominant bug class (see §9). Fixed in a641471: job-status, LIST_DIR over-capture, CREATE_FILE dir capture, multi-command banter. ****Refactor note:**** the file is a long if-ladder; a route-table + per-route unit fixtures (`tests/router_test_data.json`)

already exists) would make ordering bugs (like the job-vs-list precedence) visible.

4.2 Executor — `execution/executor_enhanced.py` (14340) Implements ****174**SUPPORTED_ACTIONS(:18** Each action is an if a == "X": block returning {ok, action, content, response, ...}. Security-gated file writes (`CREATE_FILE:8829` via `runtime/security.SecurityManager.is_path_allowed`). Background work offloaded via `runtime/background_tasks.py` (`CHECK_JOB/BACKGROUND_JOBS` at:8679 ****Status:**** largest file in the repo; the action-dispatch is a giant linear scan. ****Refactor note:**** a dispatch dict {ACTION: handler}' would cut the scan and make the action surface

enumerable for docs/tests.

4.3 Kernel / engine — `kernel/engine.py` (13374) **CognitiveEngine** owns the pipeline, mode contracts, grounded-control synthesis, and the reasoning-strategy runners (`_run_chain_of_thought:6131`, `_run_tree_of_thoughts:6223`, `_run_self_consistency:6506`). Grounded synthesis helps `_compact_grounded_synthesis(≈12946)` and `_synthesize_answer(13118)`. ****Status:**** second-largest file; very high coupling. The empty-‘cascade originates here (synthesis call shapes) — see governor

doc. **Refactor note:** the synthesis/think/budget logic should move behind the Adaptive Inference Governor (companion doc) rather than per-site flags.

4.4 Agent bus — `cognition/agent_bus.py` (3138) ****14 agents**** (verified classes): `memory(723)`, `system`. Profiles selected per action by `_select_profile` (default branch + keyword heuristics; this is what dropped `system` on the compound `CREATE_FILE` turn — §9). Dependency DAG: `knowledge_graph ← memory(_AGENT_DEPENDENCIES,:596)`, executed in topological layers via `core/dag.py`. ****[V]**** ****Note [B]:**** memory index records `FrontierAgent` as dormant and bus agents as

retrieval-level rather than frontier-quality — not re-verified this session.

4.5 Inference — `cognition/gguf_inference.py` (2677) + `inference_broker.py` `llama.cpp` wrapper. Model-agnostic resolution (`get_model_path`), family-aware templating (`_is_chatml_model/_is_llama_model/_is_mistral_model`, filename-based), no-think prefill (`_no_think_prefill`), thinking-model detection (`_is_thinking_model:219`, filename allowlist), live decode-speed EMA (`record_speed`, non-stream path:1162). All calls serialize on `LLM_CALL_LOCK`. ****Status:**** the think/budget policy is incomplete for cross-hardware use — full analysis + plan in `inadaptive_inference_governor.md`. `force_no_think()` added in `a641471`

(one caller). **[V]**

4.6 Memory — `memory/memory.py` (4461) + `memory/4` SQLite stores + FTS5 mirrors + FAISS + KG (§5). Hybrid recall: SQLite LIKE + FTS5 + FAISS + KG, merged/reranked (`cognition/reranker.py`, `cognition/scoring.py`). Entry classes `Memory`, `get_memory()`, `recall_memory()` **[L/V]**

Note [B]: memory index records 3 memory-ranking fusion weight sets that still differ (deferred tuning) — not re-verified.

4.7 Perception — perception/ (20 files)

- **Vision** `vision.py`: model-agnostic VLM via `llama.cpp` multimodal; `mtmd` encoder forced to CPU to avoid a CUDA clip segfault (:459). **Documented default Qwen2.5-VL; runtime in the log resolved Moondream2** (`MoondreamChatHandler`) — a discrepancy (§9). OCR path exists (`analyze_image.py`, `tesseract`) but “read the text” routed to captioning in the log. **[L/V]**
- **STT** `audio_stt.py` (1848): faster-whisper `small.en`, CPU, int8 (*runtime snapshot*). **[L]**
- **TTS** `tts_router.py`: Piper-only, `en_US-amy-medium`, `aplay`. **[L]**
- Also: screen read/analyze, gaze engine, ambient vision daemon, PDF/CSV analysis.

4.8 Coding agent — coding/ (12 files, 2,031)

Self-contained frontier coding subsystem: plan → DAG/tree search → execute → verify → repair (`coding/__init__.py` docstring **[V]**), sandbox at `coding/sandbox.py`, `plan_graph.py`, `cost.py` (`should_background`). Invoked by `CODE_SOLVE` and by self-improvement. **[V] Note [L]:** the coding agent’s fix-generation calls hit the same empty-‘<think> failure (`SELF_IMPROVE` turn) — covered by the governor plan.

4.9 Planning / proactive — planning/ (24 files)

proactive_daemon.py (1301): continuous habit/pattern learning, persona-overlay refresh, autonomy tick (code-monitor + self-model + scheduler proposals), news synthesis. Also jobqueue.py (durable subprocess jobs — distinct from in-process runtime/background_tasks.py), schedulers, and the world-room avatar. **Status [L]:** the daemon emits very frequent persona_updater/kg sync churn (§9 G). **[V/L]**

4.10 Learning / self-improvement — learning/ (12) + runtime/self_improvement.py (1206) Self-model refresh, session summaries, habit detection/rules, failure analysis (**SELF_ANALYZE**), fix proposals (**SELF_IMPROVE**→ coding agent, propose-only). **self_improvement.py:641**uses **broker.infer(max_tokens=700)**(small → already no-think). ****[V]**** ****Note [B]:**** memory records **profile_extractor** as a regex→canned-phrase table with a hard-coded **active_project="developing ELI"**, and **time_habit'** as stale/unwindowed — not

re-verified this session.

4.11 Core — core/ (23 files)

Hardware + runtime substrate: hardware_profile.py (1,040, free-VRAM sizing, compute reserve :69), startupHardware_optimizer.py (boot profile, train_ctx_for_model:164), model_tier.py (size tier + **live tok/s EMA**), runtime_settings.py (1,034, portable storage / path healing), paths.py/portable_paths.py/legacy_paths.py/db_paths.py, netguard.py (204, offline-by-default socket failsafe), dag.py (474, orchestrator), crisis_guard.py, full_control.py, model_download.py, cognition_tunables.py. **Finding [V]:** dynamic_runtime_budget.py (DynamicRuntimeBudget) has **zero references repo-wide** — a dead/unwired module that is a half-built hardware-budget the governor could fold in (§10).

4.12 Runtime — runtime/ (79 files, 22,225)

The catch-all: deterministic_grounding_gate.py (4292), grounded_remediation.py (1604), generated_script_guard.py (1072), control_contracts.py (943), security.py (207), approval_engine.py (103), background_tasks.py, command_splitter.py, evidence/identity/ operator/response sub-families. **Refactor note:** this package is the least cohesive; many single-purpose guard/contract files. A pass to group by concern (grounding / guards / evidence / contracts) would improve navigability.

4.13 GUI — gui/ (6 files, 17,618)

PyQt app eli_pro_audio_gui_MKI.py (10,985) + labs_tab.py (5696) + panels. 12 main tabs **[B]** (Report Builder promoted; Proactive×6 / Labs×10 / Settings×5 sub-tabs per memory index). QScintilla optional (falls back to basic editor). **Status:** large monolith GUI files.

4.14 World / plugins / tools / cli

- world/ — avatar “rooms” (Reflection Chamber, Memory Archive); gated to world/activity queries (a greeting leaked a room narration in the log — §9). **[L]**
- plugins/ — plugin host (3 files).
- tools/ — news/news_fetcher.py (RSS/arXiv/HN/Reddit), image_engine/ (SSD-1B + procedural).
- cli/ — headless REPL pieces; eli/__main__.py --headless, --trust-agent.

5. Data & state stores (table counts: runtime snapshot 2026-06-17)

Path: artifacts/db/. **[L]** (from EXPLAIN_MEMORY_RUNTIME)

Store	Size	Tables	Role
user.sqlite3	58.32 MB	38	the real store: memo-ries, conver-sa-tion_turns, kg_entities/relation news, habits, observa-tions, re-call_log, run-time_events, ses-sion_summaries, work-ing_memory_pins, + FTS mirrors
agent.sqlite3	2.66 MB	25	agent_dispatches, agent_metrics, code_patches, failures, improve-ments, observa-tions
system_index.sqlite3	0.566 MB	4	desktop_apps, executa-bles, re-cent_files, user_dirs
coding_memory.sqlite3	0.02 MB	1	coding_bug_fixes

- **FTS5 mirrors:** memories_fts, kg_entities_fts, news_fts.
- **FAISS:** artifacts/vectors/index.faiss (+meta.json), 1,228 vectors, dim 768, embedder models/embeddings/nomic-embed-text-v1.5.Q4_K_M.gguf.
- **KG:** kg_entities (50) / kg_relations (49).
- Live counts at snapshot: memories 1,265; conversation_turns ~4,426; distinct sessions 967; learning_replay 6,189; runtime_events 16,426; news_articles 1,801.
- **Note:** “active/user/memory DB roles alias the same user.sqlite3 file” — 4 *physical* files, more logical roles. **[L]**

6. Action / capability surface

- capability_manifest.json — **208** capabilities (regenerated at startup; generated_at 2026-06-17). capability_inventory.generated.json is the generated inventory.
- executor_enhanced.SUPPORTED_ACTIONS — **174** action strings (:1899). The gap between 208 capabilities and 174 actions is expected (capabilities include GUI/registry items that aren’t executor actions). **Note:** the maintainer-memory figure “155 actions / 194 caps” is **stale**; the verified current

numbers are 174 / 208.

- Full human catalogue: blueprints/capabilities_and_actions.md, capability_catalogue.md. **[B]**

7. Reasoning modes & inference policy

- Modes (cognition/reasoning_modes.py): quick, normal, advanced, research, expert (visible names), plus strategy runners CoT / ToT / self-consistency / constitutional. **[V]**
- Per-mode budget (:312–352): derived from n_ctx, prompt pressure, query complexity, mode — **no throughput term**. Pass-count (not length) is capped by measured tok/s via model_tier.speed_passes() (engine :6241, :6520). **[V]**
- **This is the active work area**. The full analysis (think/no-think, throughput-blind budgets, filename-based capability detection, empty-<think> recovery, the unused DynamicRuntimeBudget) and the proposed **Adaptive Inference Governor** are in blueprints/adaptive_inference_governor. Treat that as the authority for this layer.

8. Cross-cutting invariants (must hold across all changes) [B, maintainer policy]

1. **Model-agnostic** — no hard-coded model name/size on the inference path; frontier-quality bar.
2. **No fake actions** — if ELI says it did X, the pipeline must actually do X (no narrated success).
3. **Offline-by-default** — all internet via core/netguard; process-wide socket failsafe.
4. **Redistributable / no personal data** — never hard-code a user name or /home/<user> path in source; runtime redaction exists. Settings are stored portably (_portable_settings_for_storage).
5. **Grounding** — grounded/self-referential questions must use real evidence, never CHAT-guessing; degenerate -/empty fragments are discarded, not surfaced.
6. **Single inference broker / serialized GGUF** — one model call at a time (_LLM_CALL_LOCK).
7. **Check the codebase before proposing** — features usually already exist; frame work as gap-analysis, not greenfield.

9. Consolidated known-issues register

Severity P0=integrity, P1=routing/exec, P2=quality. Status as of a641471.

#	Issue	Where	Status
R1	Job-status queries ("job #3") didn't reach CHECK_JOB (# broke regex)	router	FIXED a641471
R2	CREATE_FILE dropped target dir + wasn't executed (profile dropped system)	router + engine fast-path	FIXED a641471
R3	LIST_DIR over-capture (swallowed trailing clause; "my notes"→dir)	router	FIXED a641471
R4	Multi-command banter folded into media title	command_splitter	FIXED a641471

#	Issue	Where	Status
R5	EXPLAIN_* compact synthesis empty-<think> loop	engine + gguf	PARTIAL (compact) a641471; governor plan for the rest OPEN → governor
I1	Catastrophic latency (2-9 min/turn) — throughput-blind budgets on slow HW	inference policy	OPEN → governor
I2	Standard-synthesis fallback + coding fix-gen still empty on <think>	engine + coding	OPEN → governor
I3	Thinking-detection is a filename allowlist	gguf_inference:219	OPEN → governor
I4	ctx table undershoots model n_ctx_train (32768 vs 262144)	train_ctx_for_model	OPEN → governor
I5	LIST_NOTES success discarded by low-grounding downgrade→CHAT (wrong “just one”)	engine	OPEN (parked)
I6	Confidence gate checks response conf, not grounding (passes grounding=0.0)	engine stage 12	OPEN
I7	Vision = captioning, not OCR; “read the text” didn’t OCR; Moondream2 vs documented Qwen2.5-VL	perception/vision	OPEN
I8	Memory-count inconsistency (3295 vs grounded 1265) across phrasings	memory introspection	OPEN
I9	Multi-part question half-answered (persona Q ignored)	engine/synthesis	OPEN
I10	World-room narration leaked into a plain greeting	world + engine gate	OPEN
I11	Proactive persona_updater/KG-sync churn (deterministic, redundant; CPU/DB, not LLM-lock)	proactive_daemon/persona_updater	OPEN → governor §G

#	Issue	Where	Status
I12 [B]	profile_extractor regex→canned, active_project hard-coded; time_habit stale; frequency-habits never become rules	learning/proactive	OPEN, not re- verified
I13 [B]	FrontierAgent dormant; bus agents retrieval-level not frontier	agent_bus	OPEN, not re- verified

10. Dead / unwired / duplication (refactor targets)

- **core/dynamic_runtime_budget.py** — DynamicRuntimeBudget has **zero references** repo-wide. Dead; overlaps the governor's intended budget object. **[V]**
- **eli/integrations/** (1 file, 0 LOC) — empty package. (eli/guards/ was the other empty package; it has been **removed**.) **[V]**
- **eli/tools/image_engine** appears both as a 1,750-LOC module and a nested image_engine/image_engine/ package (engine 1,483 + visual_core 1,447) — possible duplication to reconcile. **[V]**
- **Two inference-broker concepts** — verified the live one is cognition/inference_broker.py; core/inference_broker.py does **not** exist. Don't reintroduce a second.
- **[B] duplication convergence** (memory): canonical cognition/scoring.py exists; topic-stopwords / FTS-splits / two-tier routing are distinct (not dup); 3 memory-ranking fusion weights still differ — deferred.
- Executor (174-way if) and router (long if-ladder) are the two biggest *structural* refactor candidates (dispatch table + route table).

11. Tests & verification

- 194 test files; full run **7,347 passed / 5 failed / 45 skipped / 2 xfailed** (2026-07-01). **[V]** The 5 remaining reds are the in-progress smart_home plugin removal + one stale blueprint ref — pre-existing, unrelated, fail identically on a clean tree.
- Groups: tests/claims/ (project-vs-claims, symbol inventory, agent contracts), tests/regression/, tests/generated/ (+ _manifest.json), plus the flat test_* suite.
- No-GGUF pattern: many tests assert engine/router behaviour without loading a model (tests/test_*_no_gguf) — the right place to add governor decision tests.
- Isolation via in-project ELI_ARTIFACTS_DIR (no monkeypatch). **[B]**
- Report: tools/run_test_report.py → artifacts/test_report.md; run_tests.sh.
- **Baseline at a641471:** router/multi-command/background (239) + filesystem/think-stop/reasoning/runtime-status (61) green; full collection clean. **[V]**

12. Install / packaging / redistribution

- install.sh (+ .bat/.ps1): frozen lock (requirements.lock.txt) + GPU verify + DB init. **[B]**
- Platform requirement files: requirements-{windows,macos,android,full,learning}.txt.
- pyproject.toml, eli-mkxi.desktop, build_packages.sh, packaging/.
- Redistribution mechanics: portable settings storage + stale-path healing (runtime_settings._heal_model_paths, _resolve_relative_model_paths), dataset redaction. **Open risk [B]:** personal values (e.g. user_name) can still end up tracked; the cross-platform generalization mandate (Linux/macOS/Windows) is ongoing.

13. Blueprint index (existing design docs in blueprints/)

architecture.md, architecture_ascii.md, agent_algorithms.md, orchestration_and_agents.md, dag.md/dag_orchestrator.md, memory.md, grounding_and_evidence.md, inference_and_hardware.md, coding_agent.md, code_mode_execution_layer.md, learning.md, perception.md, gui.md, security.md, operations.md, installation.md, capabilities_and_actions.md/capability_catalogue.md, what_eli_can_do.md/what_eli_is.md, project_overview.md, state_snapshot.md, runtime_planning_world.md, eval_harness.md, lora_pipeline.md, background_tasks.md, model_bakeoff_dossier.md, top2_head2head.md, decomposition_plan.md, proposal_total_awareness.md, complete_findings.md, diagrams.md. + **this audit** and **adaptive_inference_governor.md**.

14. Appendix — verification provenance

Read this session [V]: router (route, job/list/create_file routes, _expand_common_dir), command_splitter, engine(_PHASE45_DIRECT_FAST_ACTIONS, _run_chain_of_thought, _compact_grounded_synthesize_answer, replan, low-grounding downgrade), gguf_inference (_no_think_prefill, _is_thinking_model, record_speed, broker legacy paths), reasoning_modes budget, model_tier (EMA + speed_passes), hardware_profile(reserve), startup_hardware_optimizer.train_ctx_for_model, agent_bus (14 agents + _select_profile branches), persona_updater, dynamic_runtime_budget (dead), executor CREATE_FILE / CHECK_JOB handlers, SUPPORTED_ACTIONS, structure/LOC/test counts. **From the 2026-06-17 session log [L]:** model load, stage map, DB table dump, vision/STT/TTS runtime, latency/empty-<think> timings. **From blueprints / maintainer memory [B], not re-verified:** GUI tab inventory, install internals, FrontierAgent dormancy, profile_extractor/time_habit specifics, memory-fusion weights, duplication-convergence status. Re-verify any **[B]** item before relying on it for a fix.